Assignment 4

Deadline: Monday, 9 November 2026, 18:00

Introduction

In this assignment, you will implement two methods for approximate inference in Bayesian networks: *likelihood weighting* and *Gibbs sampling*. Both methods approximate conditional probabilities using samples, but they handle evidence in different ways.

All tasks are programming tasks and are to be completed in the provided Jupyter notebooks `likelihood_weighting.ipynb` and `gibbs_sampling.ipynb`. The notebooks contain the detailed function signatures, return types, examples, and test cases.

Submission instructions:

- Submit **the completed Jupyter notebooks (`.ipynb`)** via Moodle. Submit **notebooks only**: no solution slides, separate PDF, or written answers are required.

Do not forget to:

- indicate which parts of the assignment you have completed here and here;
- select an interview time slot here;
- provide feedback here.

1 Likelihood weighting (4 points)

Complete the notebook `likelihood_weighting.ipynb` by implementing the functions `sample_lw` (part 1.1, 2 points) and `likelihood_weighting` (part 1.2, 2 points).

2 (Bonus) Gibbs sampling (4 points)

Complete the notebook `gibbs_sampling.ipynb` by implementing `resampling_distribution` (part 2.1, 2 points), `sample_gibbs`, and `gibbs_sampling` (part 2.2., 2 points).